

# OTIC Studio

## Product Requirements Document (PRD)

**Version:** 1.0

**Owner:** OTIC Technologies

**Status:** Foundational Product Specification

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## 1. Product Vision

OTIC Studio is a fully offline AI-powered Learning Operating System designed to help learners:

```
graph TD; A[Understand] --> B[Practice]; B --> C[Apply]; C --> D[Create]; D --> E[Collaborate]; E --> F[Teach Others];
```

Understand  
↓  
Practice  
↓  
Apply  
↓  
Create  
↓  
Collaborate  
↓  
Teach Others

The platform combines:

- Educational Application
- AI Tutor
- Learning Operating System
- Education Platform

into a single ecosystem.

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## 2. Mission

Enable anyone to learn anything, anywhere, completely offline.

OTIC should empower learners to:

- Understand concepts deeply

- Apply knowledge practically
  - Build real-world projects
  - Collaborate with others
  - Become independent thinkers
  - Teach others
- 

## **3. Core Principles**

### **Principle 1: Learning Must Lead to Action**

Knowledge alone is not enough.

Every learning journey should eventually lead to practical application.

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### **Principle 2: Every Answer Must Build Understanding**

OTIC should explain concepts clearly and appropriately for the learner.

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### **Principle 3: Learning Should Be Adaptive**

The system should adapt based on:

- Age
  - Learning style
  - Knowledge level
  - Interests
  - Progress history
- 

### **Principle 4: Learning Should Be Practical**

Whenever possible, OTIC should encourage:

- Projects
  - Experiments
  - Simulations
  - Real-world application
-

## Principle 5: OTIC Is A Mentor

OTIC should not behave like a search engine.

OTIC should:

- Guide
  - Encourage
  - Challenge
  - Coach
  - Mentor
- 

## 4. User Roles

### Guest

Purpose:

Demonstration Mode

Capabilities:

- Ask questions
- Experience AI tutoring
- Explore learning features

Restrictions:

- No saved progress
  - No certificates
  - No memory
- 

### Student

Capabilities:

- AI Tutor
- Voice Learning
- Simulations
- Learning Paths
- Quizzes
- Projects
- Achievements

- Certificates
  - Learning Memory
  - Collaboration
- 

## Teacher

Capabilities:

- View student performance
  - Monitor strengths
  - Monitor weaknesses
  - Create learning groups
  - Generate quizzes
  - Generate assessments
  - Track progress
  - Guide students
- 

## Admin

Capabilities:

- User Management
  - School Management
  - Update Management
  - Device Management
  - Platform Administration
- 

# 5. Navigation Structure

Main Navigation:

- Home
  - Learn
  - Practice
  - Create
  - Projects
  - Achievements
  - Certificates
  - Collaborate
  - Teacher Portal
  - Settings
-

## 6. Home Screen

### New Learners

Display:

- Welcome Message
  - Ask Anything
  - Explore Topics
  - Recommended Learning
- 

### Returning Learners

Display:

- Continue Learning
  - Recent Topics
  - Current Projects
  - Achievements
  - Recommended Practice
  - AI Mentor Suggestions
- 

## 7. Learning Modes

### Learn Mode

Purpose:

Understanding Concepts

Examples:

- Explain gravity
- Explain entrepreneurship
- Teach me programming

Outcome:

Conceptual understanding

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## Practice Mode

Purpose:

Reinforcement

Includes:

- Exercises
- Challenges
- Problem Solving

Outcome:

Retention and confidence

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## Apply Mode

Purpose:

Use knowledge in real-world situations

Examples:

- Budget creation
- Farming plans
- Programming exercises
- Business planning

Outcome:

Practical competence

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## Create Mode

Purpose:

Build something meaningful

Examples:

- Mobile App
- Business Plan

- Science Project
- Agriculture Project

Outcome:

Creation and innovation

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## Teach Mode

Purpose:

Mastery through teaching

Flow:

Student explains topic.

OTIC evaluates understanding.

Outcome:

Deep learning and mastery

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## 8. AI Tutor Behaviour

Every AI interaction should follow:

Answer  
↓  
Clarify  
↓  
Practice  
↓  
Apply  
↓  
Create  
↓  
Reflect

OTIC should never stop at answering.

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## 9. Learning Paths

When a learner requests:

Teach me Python

OTIC should automatically generate:

```
Introduction
↓
Variables
↓
Conditions
↓
Loops
↓
Functions
↓
Projects
↓
Assessment
↓
Certification
```

This structure applies across all learning domains.

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## 10. Adaptive Learning Engine

OTIC should detect:

- Age
- Skill Level
- Progress
- Learning Style
- Interests

And adapt:

- Language complexity
- Examples
- Activities
- Projects



accordingly.

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## 11. Gamification System

Features:

- Points
- Experience
- Skill Levels
- Achievements
- Badges
- Streaks
- Milestones

Examples:

- Science Explorer
  - Science Builder
  - Science Creator
  - Science Mentor
- 

## 12. Certification System

After successful completion:

Generate:

- Digital Badge
- Certificate
- Completion Record

Requirements:

- Offline generation
  - Printable PDF
  - Stored locally
-

## 13. Collaboration System

Students should be able to:

- Join learning groups
- Work on projects together
- Share ideas
- Participate in challenges
- Learn from peers

Collaboration must remain educational and constructive.

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## 14. Simulation Engine

Supported Domains:

- Science
- Mathematics
- Programming
- Engineering
- Business
- Agriculture
- Economics
- Financial Literacy
- Entrepreneurship

Simulation Flow:

```
Student Request
↓
OTIC Suggests Simulation
↓
Student Confirms
↓
Simulation Launches
```

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## 15. Voice Learning

Features:

- Offline Speech-To-Text

- Offline Text-To-Speech
- Voice Tutoring
- Voice Commands

Storage Policy:

Store:

- Converted Text

Do Not Store:

- Audio Recordings
- 

## 16. Student Memory Engine

Store:

- Age
- Interests
- Goals
- Learning Style
- Strengths
- Weaknesses
- Projects
- Achievements
- Progress
- Certificates

Use compressed memory.

Store summaries rather than entire conversations.

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## 17. Teacher Dashboard

Features:

- Student Overview
- Performance Tracking
- Strength Analysis
- Weakness Analysis
- Assessments
- Projects

- Recommendations

Teachers should be able to support learners effectively.

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## 18. Admin Dashboard

Features:

- User Management
  - Device Management
  - School Management
  - Update Management
  - System Monitoring
  - Analytics
- 

## 19. Emotional Safety Engine

Detect:

- Bullying
- Harassment
- Abuse
- Anxiety
- Stress
- Depression Indicators

Response Strategy:

1. Listen respectfully
2. Offer support
3. Encourage trusted adults

Recommendations:

- Parent
- Teacher
- Counselor
- Responsible Adult

OTIC is not a therapist.

OTIC is a learning companion.

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## 20. Knowledge Domains

### Academic

- Mathematics
  - Physics
  - Chemistry
  - Biology
  - Geography
  - History
  - English
- 

### Technology

- Programming
  - Artificial Intelligence
  - Data Science
  - Robotics
- 

### Business

- Entrepreneurship
  - Marketing
  - Finance
  - Sales
- 

### Agriculture

- Farming
  - Livestock
  - Agribusiness
- 

### Life Skills

- Leadership
  - Communication
  - Critical Thinking
  - Innovation
-

## Professional Skills

- Research
  - Digital Skills
  - Project Management
- 

## 21. Offline Requirements

Internet Required:

NO

Cloud Required:

NO

External APIs:

NO

Everything must work locally.

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## 22. Update Strategy

Supported Methods:

### USB Flash Drive

Manual deployment.

### Local School Server

Mass deployment.

Purpose:

Enable updates without internet connectivity.

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## 23. Device Requirements

### Android

Minimum:

- 4 GB RAM
  - 32 GB Storage
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### Windows

Minimum:

- 8 GB RAM
- 

### Ubuntu

Minimum:

- 8 GB RAM
- 

## 24. AI Model Strategy

### Phase 1

Gemma 3 1B

Benefits:

- Lightweight
  - Faster
  - Offline Friendly
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### Phase 2

Gemma 4

Benefits:

- Improved reasoning
- Better tutoring
- Advanced mentoring

Distributed as optional upgrade.

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## 25. Success Metrics

A learner should be able to:

- Understand concepts
  - Apply knowledge
  - Build projects
  - Solve problems
  - Create value
  - Teach others
  - Collaborate effectively
  - Earn certifications
- 

## 26. Long-Term Vision

OTIC Studio should become:

```
AI Tutor
+
Offline School
+
Learning Operating System
+
Education Platform
```

A platform that empowers learners to become:

- Creators
  - Innovators
  - Entrepreneurs
  - Professionals
  - Lifelong Learners
  - Future Teachers
-



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